Cloud Migration strategy :

- Identify existing applications, databases, dependencies, performance requirements, and security risks.
- select a suitable cloud provider and migration schedule.
- Move a small application first to test compatibility and performance.

(2) Cloud - Based IoT architecture for a smart city

Introduction :

A smart city uses thousands of Internet of things (IoT) sensors to monitor traffic, air quality, electricity consumption, and public safety. These sensors generate large amounts of data that must be collected, processed, and public safety.

(4) Software defined data center (SDDC):

SDDC manages and networking through software

→ compute virtualization : create and manages VM software.

→ faster providing automation

SDDC is more flexible stable and cost efficient

→ Proposed solution:

:) Docker: packages application and dependencies into containers

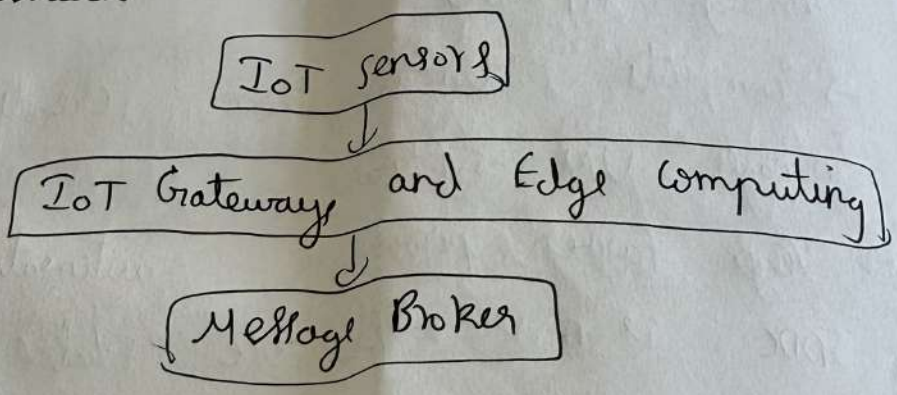
:) Manages containers development, scaling and recovery

Service Discovery and Scaling:

:) Automatically identifies and connects services using kubernets services and DNS

:) Auto Scaling: increases or decreases ntainer replicas based on work load.

Architecture :



A fintech ~~Stream~~ F startup can use containerys and orchestration to deploy applications consistently across multiple cloud providers

3) VM Criticization Analysis :

- CPU 92%
- Memory 85%
- Storage I/O 35ms

→ High utilization causes processing delay

→ memory may cause swapping and slow performance

Service discovery and scaling:

- Automatically identifies and connects services using Kubernetes service and DNS

Conclusion:

Docker and Kubernets help maintain consistent deployments, simplify service management and scale application efficiently.

Name: M.H. Yameenullah
Reg No: 192525350
Subject: Cloud Computing

Course code: CSA1519

CO2 AT-2

① Cloud Migration strategy for a Manufacturing company.

Introduction:

Cloud migration is the process of moving applications, databases, and IT resources from an on-premises data center to a cloud platform

Architecture:

